

1 Sharing sunlight between trackers and crops: dynamic
2 light-sharing control for agrivoltaics under a
3 food-security floor on the world's staple croplands

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6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift the daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a food-security floor of 90% of open-field yield with a differentiable simulator of the whole system. The crop response to shade is the least certain part: a hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, and we turned the learned curves and four field trials into ensembles of plausible responses by gradient descent through the simulator. In a deployable protocol (density chosen from each site's history, held-out sites and years, 400 crop–site pairs), light-sharing rules allowed arrays three times as dense as static arrays at equal retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24; a phenology rule needing no forecast was the best. A neural controller exceeded the written rule by 0.044 LER but a rule tuned on the same training seasons by only 0.018 (95% interval 0.012–0.024); training on the ensembles or adding a farm-measurable biomass signal did not help. The design decision mattered more: a density chosen under the mean response left about 70% of crop–site pairs at the floor when another plausible response held (oracle 85–87%), and a floor raised by 0.02 under the mean response did better at equal compliance than a rule built on the ensemble's quantiles. Under illustrative costs, light sharing breaks even with the open field at 73–92 USD MWh⁻¹ and a conventional plant replacing the crop stays more profitable. For maize and soybean (53% of the upscaled 14 PWh yr⁻¹) the direction of the simulator's bias is uncertain, and the meta-analysis points to overestimated gains.

7 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,

9 **1. Introduction**

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
33 date adaptation measure against heat and drought stress, in addition to a
34 mitigation measure through renewable electricity. The same shade, however,
35 also reduces the light available for photosynthesis. Regulators increasingly
36 require AV systems to retain most of the reference crop yield. The German
37 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
38 reference yield (DIN, 2021), and other schemes and many farmers set stricter
39 floors. How much electricity can be produced without sacrificing food secu-
40 rity depends on the crop, the local climate, the year and the growth stage.

41 Most AV designs fix the geometry of the array, i.e. the ground coverage
42 ratio (GCR, module area per ground area) and the tilt or tracking rule, from

generic rules of thumb. Single-axis trackers add a degree of freedom: the modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which is sample-inefficient and hard to scale to global, multi-decadal assessments.

Two questions follow. First, how much of the electricity that an agrivoltaic system can produce under a food-security floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-

analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but milder than the meta-analysis for maize and soybean, whereas one maize trial is milder still; the gains for those two crops are therefore uncertain and possibly overestimated (Section 3.7).
4. A learned controller adds a small gain over the best rule: 0.044 LER over the written phenology rule and 0.018 over a rule tuned on the same training seasons. Training it on ensembles of shade responses, or giving it a farm-measurable biomass signal, does not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (Section 3.8).
5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop–site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 9 points for 0.04 LER, but a floor raised by 0.02 under the mean response does better at equal compliance, so the ensemble is

- 118 useful to size a margin and not to replace it (Section 3.8).
 119 6. Under illustrative costs, light sharing breaks even with the open field at
 120 73 to 92 USD MWh⁻¹, learning is worth about 2 kUSD ha⁻¹ yr⁻¹ over a
 121 tuned rule at 90 USD MWh⁻¹, and a conventional plant replacing the
 122 crop is more profitable at every price (Section 3.9).
 123 7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland
 124 could supply about 14 PWh yr⁻¹, but only with about 10 TWp of mod-
 125 ules, several times the installed PV capacity of the world (Section 3.11).

126 2. Materials and methods

127 2.1. Global cropland sites, weather and crop calendars

128 We studied five staple crops that together cover about 670 Mha of har-
 129 vested area and 580 Mha of physical cropland: wheat, rice, maize, soybean
 130 and potato. Harvested areas (rainfed and irrigated) were taken from the
 131 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
 132 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
 133 without replacement with probability proportional to harvested area. An un-
 134 weighted average over the sample therefore approximates an area-weighted
 135 average over global production. The 400 crop–site pairs lie in 381 distinct
 136 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
 137 GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
 138 we used the winter-wheat calendar unless the coldest-month mean temper-
 139 ature was below −8 °C, in which case the spring-wheat calendar was used.
 140 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
 141 2018): global and diffuse shortwave irradiance, maximum, minimum and
 142 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
 143 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
 144 after sowing, which gives 7600 site-seasons.

145 Climate change was represented by the global warming levels +1.5, +2
 146 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point
 147 temperatures were shifted uniformly, which keeps relative humidity approx-
 148 imately constant and raises the vapour-pressure deficit. The atmospheric
 149 CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and
 150 the three warming levels. Precipitation and irradiance were left unchanged.
 151 This delta approach isolates the thermal signal of warming; its limitations
 152 are discussed in Section 4.5.

153 2.2. A differentiable agrivoltaic simulator

154 The simulator advances all site-seasons in parallel with a daily time step
 155 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
 156 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
 157 that yields and electricity are differentiable with respect to the design and
 158 control variables.

159 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
 160 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
 161 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 162 on the horizontal is the difference between the two. Solar position follows
 163 the standard declination and hour-angle equations.

164 *Array geometry and PV electricity.* We consider elevated horizontal single-
 165 axis trackers with north–south axes. The axis height is 1.25 module widths,
 166 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 167 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 168 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 169 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 170 rows when the modules are horizontal, which is the stow position during field
 171 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 172 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 173 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 174 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
 175 to the sun, which minimises the shadow on the crop. Given a light-sharing
 176 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

177 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 178 ule, reduced by the fraction of the module shaded by the neighbouring row,
 179 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 180 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),
 181 with an hourly air temperature interpolated sinusoidally between the daily
 182 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 183 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 184 losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A con-
 185 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 186 energy reference E_{PV} .

187 *Validation of the PV model.* We compared the simulated specific yield of an
 188 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 189 (European Commission, Joint Research Centre, 2022), for a horizontal north–
 190 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 191 returned data). Across sites, the simulated and PVGIS specific yields were
 192 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean
 193 absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and
 194 with the different irradiance databases (NASA POWER vs. SARA2, ERA5
 195 and NSRDB). Electricity estimates are therefore, if anything, conservative.

196 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 197 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 198 fraction of sky diffuse reaching the ground is a two-dimensional isotropic
 199 sky-view factor. We pre-computed it for 25 values of g and 13 rotation
 200 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly
 201 (differentiably). The daily ratio s of the irradiance at the crop to that of the
 202 open field defines the crop’s shading. The canopy maximum temperature is
 203 lowered by $\kappa(1 - s)$.

204 κ is uncertain. Field studies under PV modules report small reductions
 205 of daytime crop or canopy temperature and little change in air temperature
 206 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 207 calibrated on air temperature do not transfer directly to canopy temperature
 208 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
 209 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 210 for all methods (Section 3.2). Nocturnal warming under the modules is not
 211 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 212 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 213 from the reduced radiation and canopy temperature. Soil water conserva-
 214 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
 215 therefore emerges from the model, but rain redistribution by the modules
 216 (Marrou et al., 2013a) is ignored.

217 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 218 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 219 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

220 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
 221 follows.

- 222 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 223 only on thermal time, with senescence accelerated by heat and drought
 224 stress. Shade therefore reduces the daily biomass increment but not
 225 leaf-area development.
- 226 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 227 treme temperature T_{ext} of the canopy maximum temperature.
- 228 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ ARID}$, where the agricultural reference index for
 229 drought (Woli et al., 2012) is computed from a daily soil-water balance
 230 with curve-number runoff, drainage and root water uptake. Irrigated
 231 fields have $f_{\text{Water}} = 1$.

232 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 233 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 234 drade et al., 1999). It also ignores the differences in shade tolerance be-
 235 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 236 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 237 reals and tuber crops initially lost less than proportionally; shading trials
 238 also show marked effects of shade on maize development (Ramos-Fuentes
 239 et al., 2023). We added a stage-dependent penalty. During a critical window
 240 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h \bar{d}$, where $\bar{d}$
 241 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 242 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 243 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 244 0.5, for which no specific curve was available) and lowest for potato (0.2).
 245 These values are assumptions that reproduce this ranking; they are not cal-
 246 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 247 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 248 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 249 makes shading around flowering more damaging than proportional. The net
 250 effect is closer to the field evidence than either assumption alone.

251 The thermal-time requirement to maturity was calibrated for every site as
 252 the mean thermal time between the GGCM sowing and maturity dates under
 253 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 254 The cultivar is thus adapted to the present climate, and warming shortens the
 255 season. Growth stops smoothly once the thermal-time requirement is met.
 256 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 257 is represented by a rainfed and an irrigated sub-field in proportion to its

258 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 259 *and* electricity are area-weighted over the two sub-fields.

260 2.3. SALS: simulation-trained adaptive light sharing

261 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 262 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 263 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 264 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 265 are the information available on day t :

- 266 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 267 terception f_{Solar} , relative soil water and the previous day's ARID;
- 268 • a *forecast* of the day's maximum temperature (relative to T_{heat} and
 269 T_{opt}), irradiance and ET_0 ;
- 270 • static information: the irrigation status of the sub-field, the array's
 271 GCR g , latitude, day of season and the crop.

272 Forecasts are perturbed with random errors, independently every day: Gaus-
 273 sian with a standard deviation of 2°C for maximum temperature, and mul-
 274 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .
 275 The physics always uses the true weather. The same forecast errors are
 276 applied to the rule-based controllers. The controller is conditioned on g .
 277 During training, each season is simulated with a GCR close to the density
 278 that is sustainable at its training site. This is the largest value on a coarse
 279 grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention
 280 of the site's training seasons at or above ρ , plus a uniform perturbation in
 281 $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to con-
 282 trol the densities that design selection will later pick. Outside the cropping
 283 season the trackers backtrack for maximum electricity.

284 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 285 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 286 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

287 which combines an exact (hinge) and a quadratic penalty for the food-security
 288 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the

289 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 290 daily steps of every season (back-propagation through time). This is the key
 291 difference from reinforcement learning with black-box simulators (Gautron
 292 et al., 2022), which estimates gradients from sampled returns.

293 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
 294 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
 295 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
 296 and evaluation. The network is fine-tuned through the simulator with Adam
 297 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
 298 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
 299 a numerically stable form, so that gradients remain finite even under extreme
 300 heat.

301 *Initialisation.* Trained from random weights, the controller converged to
 302 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
 303 the seasonal-sharing rule, because the gradient of electricity with respect to u
 304 is small and of the same sign for most days of a season. We therefore initialise
 305 the network by behaviour cloning of the phenology rule of Section 2.4: binary
 306 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
 307 1.7 million day-states visited by the rule in the training seasons, with 1500
 308 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
 309 policy away from this initialisation wherever doing so improves the objective.
 310 Consequently, the gain of SALS over the phenology rule measures what fine-
 311 tuning through the simulator adds to that rule, not what can be learned
 312 without it.

313 2.4. Deployable design selection and baselines

314 A real array has one fixed density, which must be chosen before it is built.
 315 Every method, SALS included, therefore uses the same *history-based design*
 316 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
 317 for which the method’s mean yield retention over the site’s 2001–2014 sea-
 318 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
 319 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
 320 at least 80% of the historical seasons. The historical seasons of the test sites
 321 are never used for training, and the weather of the evaluation years is never
 322 used for design. We compare four controllers:

- 323 • *AV static*: backtracking trackers without light sharing ($u = 0$);

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
 - 2: **for** step = $1, \dots, N$ **do**
 - 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
 - 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
 - 5: **for** day $t = 1, \dots, 365$ **do**
 - 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
 - 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
 - 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
 - 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
 - 10: back-propagate through the season; Adam update of ω
-

- 324 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
- 325 cropping season;

- 326 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
- 327 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
- 328 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
- 329 track the sun and shade the crop;

- 330 • *SALS*: the learned controller.

331 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)

332 are the references. Because the design rule is identical for all controllers, dif-

333 ferences between methods reflect the quality of control and the array density

334 that each controller can sustain under the food-security floor. We also com-

335 pare the controllers at identical GCR over the whole grid (matched-density

336 frontiers).

337 2.5. A shade-response model learned from field data

338 The simulator needs a crop response to shade that is supported by ob-

339 servations. We learned one from the open data of the meta-analysis of Laub

et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observations of yield relative to an unshaded control at relative shade reductions of 0–90% from 58 studies, with crop and crop type, experiment type (artificial shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$; maximum a posteriori estimation, so this is a hierarchical shrinkage regression with bootstrap intervals and not a full Bayesian model). We compared six models: proportional loss $y = 1 - s$; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5); partial pooling with covariates on $\log \beta$ (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests the prediction of a new study and not of a new observation in a known study. Intervals for the crop-type curves and covariate effects come from a cluster bootstrap over studies (150 resamples).

2.6. Shade-response uncertainty: gradient calibration, robust controllers and design

The crop’s response to shade is the least certain part of the simulator, so we treat it as an unknown with a distribution. Two parameters per crop enter the simulator: a radiation-use compensation c (growth is multiplied by $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop) and a multiplier h on the flowering-window harvest-index penalty. Because the simulator is differentiable, both can be fitted by gradient descent through whole simulated seasons to published yield–shading observations. The observations are the relative yields at 20% and 40% shading of the data-driven curves of Section 2.5 for the five crops (median, with an error equal to the width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice takes the curve of C3 cereals, which contains no rice), the field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023) (each season of each trial as a separate observation, error 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses the same procedure with a target error of 0.2 for the learned curves, which is the between-study spread of a new study (Section 3.7). An earlier ensemble, E1, was fitted in the same way to curves read by hand from the figures of

377 Laub et al. (2022) and was used to train the robust and adaptive controllers
 378 below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, and the param-
 379 eters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h) after every step, which is why
 380 a response milder than the data cannot be lowered further. We fitted 12
 381 ensemble members by randomised maximum a posteriori estimation: every
 382 member sees the observations perturbed by their errors and a prior draw
 383 perturbed by its width, and is optimised with Adam (40 steps, with the loss
 384 still falling slightly at the last step; 40 training-block seasons per crop and
 385 member; shading of the static arrays cached). The ensemble is a device for
 386 generating plausible responses and not a validated posterior: it has no cov-
 387 erage check, and for some parameters (the harvest-index multiplier of wheat
 388 and potato) its spread equals that of the prior. The design analyses use five
 389 fixed members (indices 0, 3, 5, 8 and 10 of 12), chosen in advance. The
 390 ensemble is used in three ways. All training and evaluation ran on a 4-core
 391 CPU; one training run takes about 20–25 minutes and the calibration of one
 392 ensemble about 7 minutes.

393 *Robust and adaptive controllers.* SALS-robust has the architecture and
 394 training of SALS (Section 2.3), but every training season is simulated with the
 395 response of a randomly drawn ensemble member that the controller does not
 396 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
 397 densities differ between members. SALS-adaptive additionally observes the
 398 realised biomass of the crop under the modules relative to an open reference
 399 plot, a quantity that can be measured on the farm by remote sensing or by
 400 a reference plot. A feedback rule that shares light whenever this ratio falls
 401 below a target serves as the hand-written counterpart; its target (1.0) was
 402 chosen on training-block seasons under the ensemble.

403 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
 404 largest GCR whose historical retention meets the floor. Under shade-response
 405 uncertainty we choose the largest GCR that meets the floor in at least 75%
 406 of five ensemble members, and compare it with a design under the ensemble
 407 mean and with an oracle design under the true response. Evaluation of this
 408 rule is leave-one-out: the truth is the held-out member and the design uses
 409 the other four.

410 *Stress tests.* Controllers and rules are also evaluated under four fixed
 411 responses, the proportional response, the response calibrated to the meta-
 412 analysis, the response fitted to the field trials and a harsh response (propor-
 413 tional growth and a 1.5 times larger harvest-index penalty). The proportional
 414 and harsh responses lie outside the range of the ensemble.

415 2.7. *Economic analysis*

416 To judge whether the electricity gains matter economically, we computed
 417 an annual net value per hectare for each method relative to the open field:
 418 electricity revenue minus the annuity and operating cost of the installed ca-
 419 pacity minus the loss of crop revenue (crop production costs are unchanged).
 420 The installed capacity is 2.1 g MWp per hectare (modules cover a share g of
 421 the land at 0.21 kWp m⁻²). Yield is dry matter, valued at assumed prices.
 422 A conventional ground-mounted plant (GCR 0.4) that replaces the crop is
 423 the comparator. All parameters (Table 13) are assumptions chosen to be
 424 representative, not data, and their ranges drive the conclusions; we therefore
 425 report break-even electricity prices and one-at-a-time sensitivities, and we do
 426 not monetise food security, which market crop prices do not capture.

427 2.8. *Evaluation protocol, statistics and upscaling*

428 Sites were split spatially: 10°×10° blocks were randomly assigned to train-
 429 ing (70%) or test (30%), so that no test site has a training site in the same
 430 block. Training uses the 2001–2014 seasons of the training sites. Design
 431 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
 432 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
 433 four climates.

434 We report the following metrics:

- 435 • the mean electricity (MWh ha⁻¹ yr⁻¹) and the relative electricity e ;
- 436 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 437 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
 438 compliance (share of sites whose mean test-season retention is $\geq \rho$);
- 439 • the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- 440 • the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

441 Statistical comparisons use site means as the unit of analysis. We report
 442 paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confi-
 443 dence intervals of the mean difference. SALS was trained with three random
 444 seeds on the main split, and with one seed on each of two further spatial
 445 splits.

446 For the global estimate, every site is evaluated in the split in which it is
 447 held out. The mean electricity and yield per hectare of each crop are scaled

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

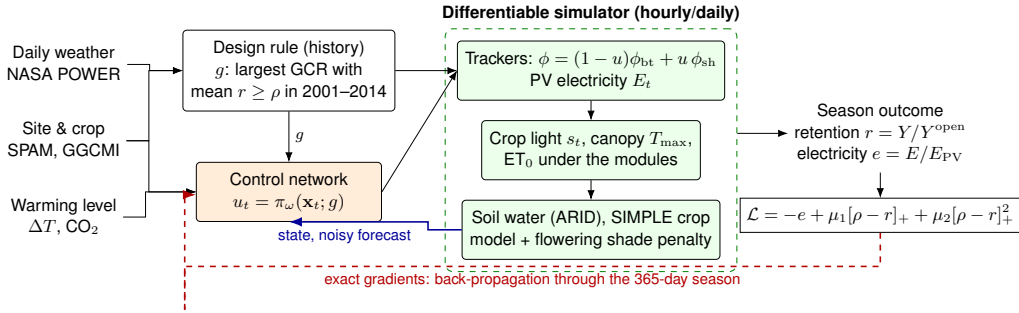


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

448 to the *physical* area of that crop in SPAM 2010, which avoids double counting
449 multiple-cropped land (Waha et al., 2020). We report the installed capacity
450 (TWp), the capacity factor and a supply curve.

451 3. Results

452 3.1. Electricity and food on held-out sites and years

453 Table 2 compares the four controllers on the 92 held-out sites of the
454 main split (seasons sown in 2015–2019), after each site’s density has been
455 chosen from its 2001–2014 seasons with the same rule for all controllers. By
456 construction all methods keep the mean yield retention close to 90% (0.904–
457 0.917), so differences in electricity are differences in the density that each
458 controller can sustain under this floor.

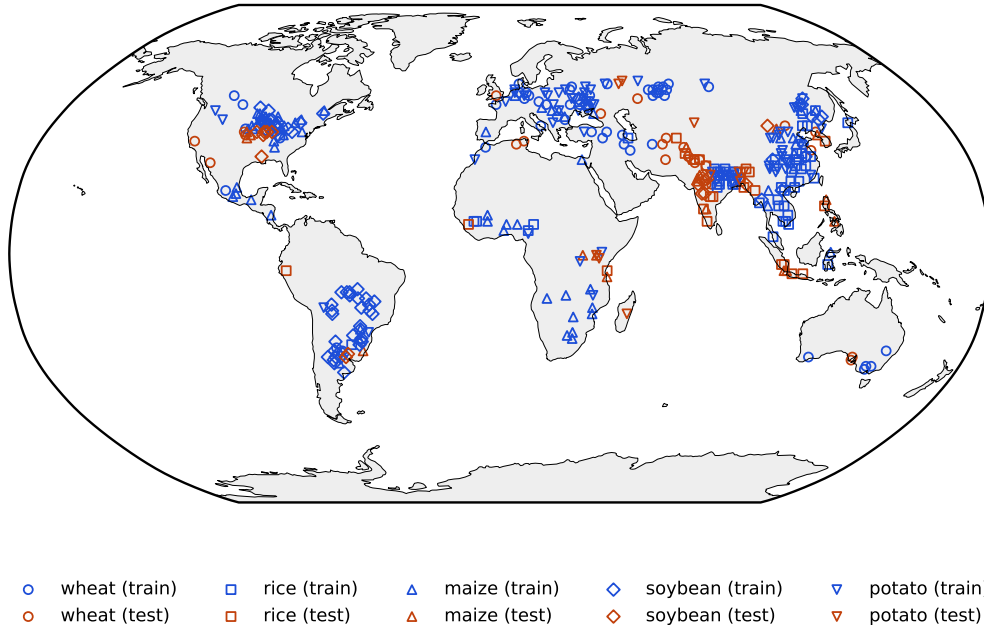


Figure 2: The 400 crop-site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

459 **Static arrays are the weakest.** Without light sharing, the floor limits
460 the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$,
461 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
462 modules to let light through during the cropping season (seasonal shar-
463 ing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule
464 that shares light only while the crop is growing (phenology rule) reaches
465 $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the
466 crop on forecast heat or drought days (stress rule) is worse than the phenol-
467 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at
468 times when the floor is most binding.

469 **The learned controller is best by a modest, consistent margin.**
470 SALS sustains a density of 17% (similar to the phenology rule), produces
471 $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Com-
472 pared with the phenology rule this is +0.038 in LER (95% CI over sites
473 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-

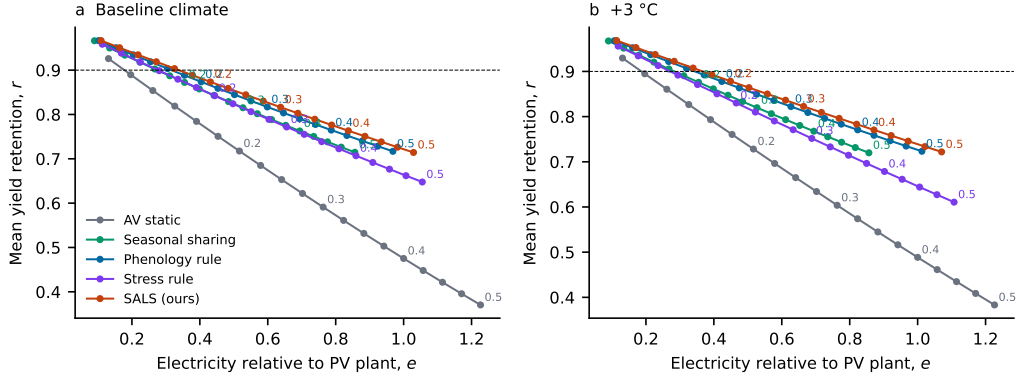


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

ble A.16). Compared with static arrays it is $+0.205$ (0.176 – 0.236), with 2.35 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows the same picture at matched densities: the learned policy lies on or slightly above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at $+3$ °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at $+3$ °C (main split: 0.034 and

0.032). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with $\kappa = 0, 1$ and 2°C . Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at 0.905 by the design rule and the ranking is unchanged. SALS exceeds the phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at $+3^\circ\text{C}$).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET₀ forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at $+3^\circ\text{C}$). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

3.3. Floor conditioning, chance-constrained training and forecast errors

One network for all floors. Regulations and farmers set different floors. The network of the main experiments was trained at $\rho = 0.9$ only. We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical design step for each floor (Table 3). Its LER exceeds that of the phenology rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037, 0.032 and 0.033 at $+3^\circ\text{C}$), so the benefit carries over to other floors without retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER ≤ 0.001). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

Chance-constrained training. We also trained a network whose objective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 ($+3^\circ\text{C}$) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

Forecast errors. We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and

of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The stress rule, which depends on the forecast of maximum temperature, degrades with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and its LER from 1.432 to 1.418 as the errors triple.

A tuned rule. The phenology rule above was written, not tuned, whereas SALS is trained. We therefore tuned the window of the rule on the training-block seasons with the same objective (electricity at the density where mean retention equals the floor); the best window starts at 0.15 instead of 0.05 of the thermal-time requirement (Table 4). Pooled over three splits and 76 distinct crop–site pairs, the tuned rule raises LER over the original rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037), and SALS exceeds the tuned rule by 0.018 (0.012 to 0.024), or 5.7% more electricity at equal retention, and the original rule by 0.044 (0.030 to 0.060). Tuning the rule thus accounts for about 60% of the margin of SALS over the written rule, and the remaining margin is small but not zero. SALS meets the floor in fewer seasons than the rules (83% against 84–85%).

3.4. *What the learned policy does*

Averaged over the rainfed test seasons of the main split at a density of 0.3, the light-sharing level of SALS is a smooth function of relative thermal time. It is close to zero in the first 10% of the thermal time required for maturity, rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and falls through 0.58 at 80–90% to about zero at maturity. The plateau covers the window in which our crop model penalises shading through the harvest index. The ramps at both ends let the modules produce electricity when additional light is worth least to the crop. Fig. 4 shows two example seasons. The phenology rule, which is also the initialisation of the network, shares light during the whole growing period. Fine-tuning through the simulator found that sharing can be reduced at the edges of the season: SALS gains its advantage by yielding light when it matters and collecting electricity when it matters less.

The policy also shares less light on days with a forecast heat stress: the mean level is 0.35 on such days against 0.60 (baseline) and 0.51 (+3 °C) on other days, and 58–63% of sites show a lower level on heat days. This cannot be read as a stress response, however. Heat days occur late in the

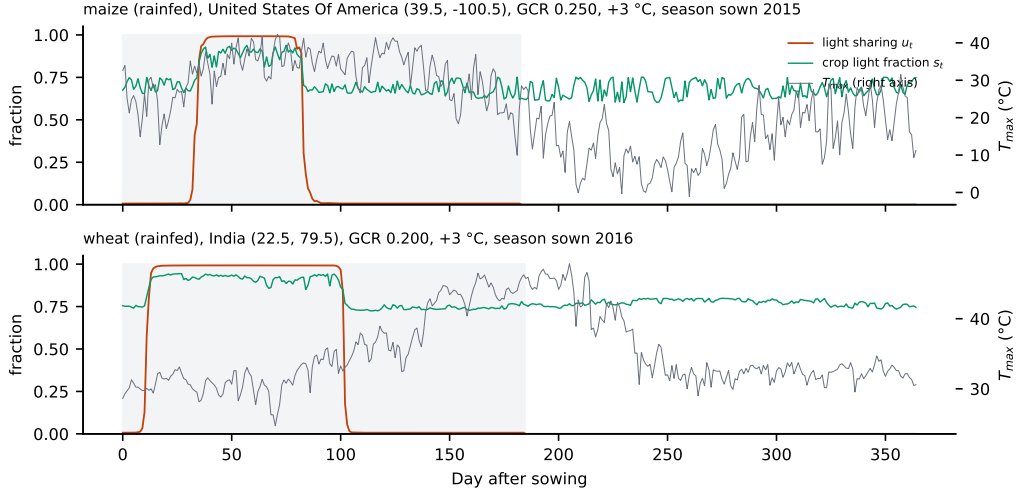


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

569 season, when the phenological ramp is already falling, and a policy without
 570 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
 571 The ablation above confirms that stress information does not improve the
 572 result.

573 3.5. Crops and regions

574 The benefit of light sharing differs between crops (baseline climate, SALS).
 575 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
 576 577 $\text{MWh ha}^{-1}\text{yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
 577 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
 578 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
 579 1.14–1.21 for the others).

580 The floor is not met evenly. Production-weighted retention is 0.888 for
 581 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
 582 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
 583 combine the highest density (GCR 0.41) and electricity ($1090 \text{ MWh ha}^{-1}\text{yr}^{-1}$)
 584 with a production-weighted retention of only 0.862. Africa and the Middle
 585 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
 586 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902

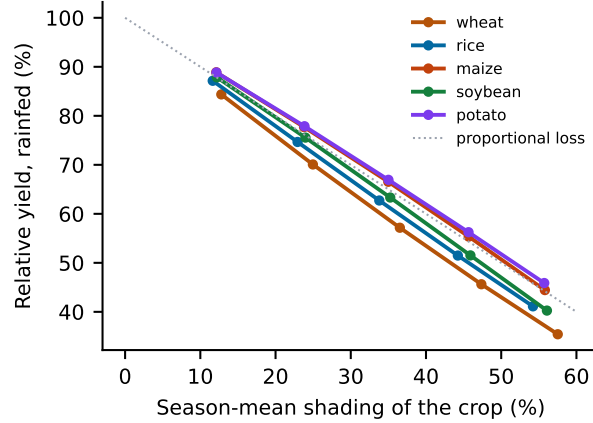


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

587 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would
 588 therefore need a region- and crop-specific design step, not a global one.

589 3.6. Model behaviour and validation

590 **PV output.** The simulated specific yield of an isolated tracker row agrees
 591 with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29
 592 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation
 593 of 5.5% (Section 2.2).

594 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
 595 crops under static arrays as a function of the season-mean shading of the crop.
 596 Yield declines close to proportionally with shading: 84–89% relative yield at
 597 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat
 598 loses most and potato and maize least. This response is steeper than the *less*
 599 *than proportional* losses that Laub et al. (2022) report for small shade levels
 600 of C3 cereals and tubers, and it does not reproduce their finding that maize
 601 and grain legumes are more shade sensitive than C3 cereals. In the simulator,
 602 shading also relieves heat and drought in rainfed maize and soybean, which
 603 offsets the higher harvest-index penalty we assigned to them. The absolute
 604 effect of shading on each crop is therefore uncertain (Section 4.5).

605 **Comparison with field trials.** We fitted one radiation-use compen-
 606 sation coefficient per crop to the yield–shading curves of Laub et al. (2022)

607 using training-block sites only (Methods, calibration script in the repository).
 608 The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the
 609 proportional model already lies within about 0.02 of the C3-cereal and tu-
 610 ber curves, and it cannot lower the maize and legume responses below the
 611 proportional case (root-mean-square error 0.16 for maize and 0.08 for soy-
 612 bean). Against independent field trials (Table 5) the model underestimates
 613 retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and
 614 potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al.
 615 (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91
 616 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was
 617 not affected by shade. The model is therefore conservative for food reten-
 618 tion with respect to these trials, except for the stronger maize losses of the
 619 meta-analysis. Table 6 shows the consequence for the design results: with
 620 the calibrated response the allowed density and the LER of every method
 621 rise (phenology rule 1.28 against 1.23), and with a response fitted to the
 622 field trials the LER of the phenology rule is 1.62. The phenology rule is the
 623 best method in all three cases, and light sharing adds LER +0.17, +0.19 and
 624 +0.30 over static arrays (conservative, calibrated, field-fitted); only the order
 625 of the seasonal and stress rules changes. The absolute gains for rice, wheat
 626 and potato are thus conservative with respect to these trials; for maize and
 627 soybean the meta-analysis points the other way (Section 3.7).

628 *3.7. A shade-response model learned from field data*

629 Cross-validation by study shows what the data can and cannot support
 630 (Tables 7 and 8). Over all 339 shaded observations the root-mean-square
 631 error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37
 632 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes
 633 from crop types that the simulator does not represent (berries, forages, fruits
 634 and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses,
 635 the learned curves are not better than proportional loss: the error is 0.222
 636 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$),
 637 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes
 638 and tubers. Partial pooling is as good as separate curves (0.312 against 0.313)
 639 and the covariates add almost nothing (0.311). Gradient-boosted trees on the
 640 same inputs are worse (0.321) than the parametric model. The error of 0.22
 641 for the four crop types is large: yields of a new study differ from the learned
 642 curve by about 20 percentage points, which is the between-study spread
 643 that a design under uncertainty has to cover. Crop type is the information

that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45). The estimated effect of solar-panel shade is not interpretable, because all 46 panel observations are of leafy vegetables, mainly lettuce, and the effect is confounded with crop type.

The learned curves show a clear ordering of crop types (Table 9). At 20% shading C3 cereals and tubers lose less than proportionally (relative yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40% shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The simulator’s responses are close to the data for cereals and potato but, under both calibrated ensembles, too mild for maize and soybean at 40% shading (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration cannot lower the maize response further, because the simulated shading also relieves heat and drought stress and the compensation and harvest-index parameters are bounded below at zero. The field trials of maize (Ramos-Fuentes et al., 2023) are also milder than the meta-analysis. For maize and soybean the direction of the bias is uncertain, and the gains reported in this paper are possibly overestimated.

3.8. Learning under shade-response uncertainty

Calibration. Gradient descent through the simulator fitted the five crops to the data-driven curves and the field trials (ensemble E2, Table 10). The data identify two features: the harvest index of maize is hardly affected by shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al. (2023), and soybean is more shade sensitive than the cereals (compensation 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the largest compensation (0.91 ± 0.31) and the largest uncertainty; the compensation is an effective parameter that absorbs diffuse-light and light-saturation effects that the simulator does not model explicitly, not a physical value.

Learned controllers. Controllers were evaluated under four fixed responses on 21 crop–site pairs of split 0 with one training seed, so the intervals below, which are cluster bootstraps over pairs of per-pair mean LER differences, do not include seed or split variation (Table 11 gives season means, which differ slightly). Under every response SALS, trained on the proportional response, is better than the written phenology rule: by $+0.019$ (0.007 to 0.039), $+0.024$ (0.009 to 0.045), $+0.038$ (0.017 to 0.063) and $+0.015$ (0.000

681 to 0.036) LER under the proportional, calibrated, field-fitted and harsh re-
 682 sponses. Training on the narrow ensemble E1 did not change this: SALS-
 683 robust differs from SALS by -0.005 to $+0.003$ (intervals contain zero except
 684 for the proportional response, 0.000 to 0.003), and SALS-adaptive, which
 685 sees the measured biomass ratio, by -0.010 to $+0.007$ (all intervals contain
 686 zero). Training on the wider ensembles did not help either: SALS-robust on
 687 E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014 , 0.017 and
 688 0.017 LER under the proportional response, 0.017 , 0.021 and 0.020 under the
 689 calibrated response, 0.004 , 0.005 and 0.011 under the field-fitted response (in-
 690 tervals contain zero) and 0.008 , 0.014 and 0.017 under the harsh response;
 691 the intervals exclude zero in 6 of the 9 comparisons for the proportional, cal-
 692 ibrated and harsh responses. The E3 robust controller is not distinguishable
 693 from the written rule ($+0.001$ to $+0.003$, intervals contain zero) except un-
 694 der the field-fitted response ($+0.032$, 0.001 to 0.067). With one seed and 21
 695 pairs these controllers therefore did not retain the margin of SALS within our
 696 training budget; a controller that must perform across plausible responses is
 697 less specialised, but a larger budget or more seeds could change this. The
 698 feedback rule is worse than the written rule by 0.026 – 0.058 LER under every
 699 response (intervals exclude zero). A plausible reason is that the biomass ratio
 700 does not reveal the harvest-index loss from shading around flowering, but we
 701 did not test this explanation.

702 **Design under uncertainty.** The larger risk is in the design step (Ta-
 703 ble 12, Fig. 6). Pooled over the three spatial splits (76 distinct crop-site
 704 pairs), a density chosen under the mean response of E2 leaves 69% of the
 705 pairs at the floor when the true response is another plausible one, against
 706 87% for an oracle design; with the wider ensemble E3 the figures are 70%
 707 and 85%. Choosing the density so that the floor is met in at least 75% of the
 708 plausible responses raises compliance by $+9.0$ points under E2 (95% cluster-
 709 bootstrap interval 5.3 to 12.9) and $+8.7$ under E3 (4.2 to 13.7), at a cost of
 710 0.038 and 0.034 in LER; it recovers about half of the gap to the oracle. The
 711 ensemble-based rule is, however, not better than a simple margin. Keeping
 712 the density rule of the main analysis and raising the floor used in the design
 713 from 0.90 to 0.92 under the mean response gives 89% (E2) and 85% (E3) of
 714 the pairs at the floor at LER 1.434 and 1.468 , whereas the ensemble rule at
 715 90% reliability reaches 89% and 88% at 1.412 and 1.409 : at equal compliance
 716 the margin keeps 0.02 (E2) to 0.04 (E3) more LER. The ensemble is thus
 717 useful to size the margin (about 0.02 to 0.04 above the floor for 85–95% of
 718 the pairs) but a rule built on its quantiles does not beat it. On the first split

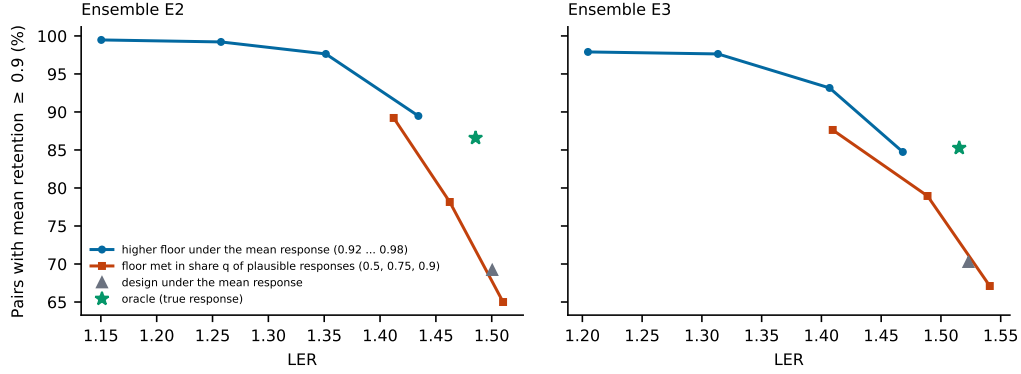


Figure 6: Share of crop–site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue), a floor met in a share q of the plausible responses (orange), the design under the mean response and the oracle design.

719 alone, with the earlier ensemble E1, the quantile rule restored the oracle level
 720 (76% and 74%, against 58% and 55% under the mean response), but with-
 721 out the margin comparison. The guarantee holds only within the range of
 722 responses that the ensemble spans: with the E2 design on split 0 (21 pairs),
 723 compliance under the proportional, calibrated and harsh responses is 19–24%
 724 (mean-response design 19%), and 81% under the field-fitted response (71%).

725 3.9. Economics of light sharing

726 All figures in this section follow from the assumed parameters of Table 13
 727 and should be read as a sensitivity analysis, not a forecast. The economics
 728 are dominated by the capacity installed, not by the crop (Table 14): the
 729 annuity of the modules is 11–32 kUSD per hectare and year, whereas the
 730 crop revenue lost is about 0.3 kUSD, so a method is worth what it earns per
 731 installed watt. With the densities designed for the floor, static arrays earn
 732 +0.3 kUSD ha⁻¹ yr⁻¹ at 60 USD MWh⁻¹ and break even at 59, whereas all
 733 light-sharing methods lose money at that price (–3 to –10 kUSD) and break
 734 even at 70 (stress rule), 73 (SALS), 77 (tuned phenology rule), 80 (written
 735 rule) and 92 (seasonal sharing) USD MWh⁻¹. At 90 USD MWh⁻¹ static
 736 arrays (6.3 kUSD) are better than the written (4.0) and tuned (5.5) rules
 737 and worse than SALS (7.6), which overtakes static arrays above about 85
 738 USD MWh⁻¹. A conventional plant that replaces the crop breaks even at 48
 739 USD MWh⁻¹ and is more profitable than every agrivoltaic option above that

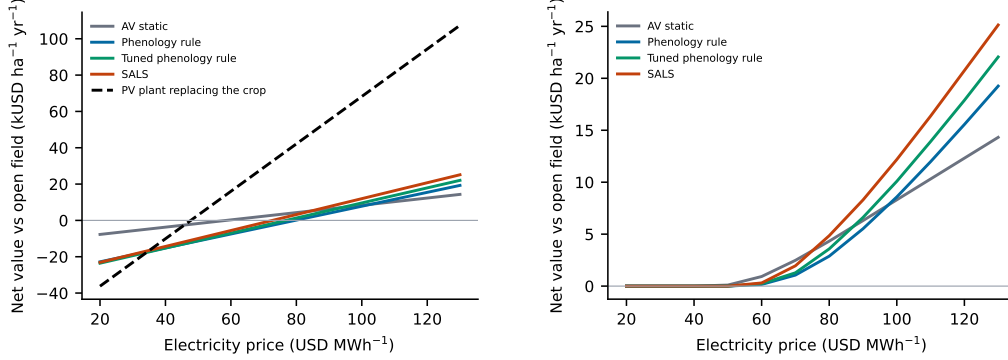


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 13.

price (16 kUSD at 60, 94 kUSD at 120, against -6 and 21 for SALS); below it all options lose money and static arrays lose least (at 30 USD MWh⁻¹: plant -23.2 , SALS -18.7 , static -5.7 kUSD). The break-even price of SALS exceeds that of the plant by about 25 USD MWh⁻¹, which is the order of the premium or the value of food production that agrivoltaics needs for parity with the open field; parity with the plant needs more. The sensitivity is large (Table 15): at 90 USD MWh⁻¹ the net value of SALS changes from $+17$ to -8 kUSD between module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30 – 40 USD MWh⁻¹; at 60 14 – 59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9 , SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh⁻¹, SALS leads from about 80 (4.9 against 4.3 at 80 , 8.3 against 6.3 at 90 , 20.7 against 12.3 at 120), and the tuned rule overtakes static arrays only above about 105 . The value of control is therefore real but depends on a high price: at 90 USD MWh⁻¹ SALS is worth 1.7 kUSD ha⁻¹ yr⁻¹ over the tuned rule with profit-optimal densities.

Cost of the safety margin. At 90 USD MWh⁻¹, moving from a density designed under the mean response to a floor of 0.92 under that response costs

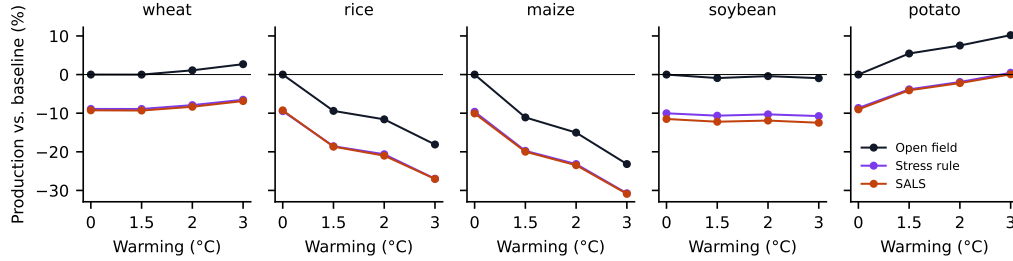


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

1.07 kUSD ha⁻¹ yr⁻¹ (18%) under E2 for 20 more points of compliance (69% to 89%) and 1.08 kUSD (18%) under E3 for 14 points (70% to 85%), about 50–75 USD ha⁻¹ yr⁻¹ per point. At 60 USD MWh⁻¹ a margin has a negative cost, because every module beyond break-even loses money; the economics of the design margin therefore depend on the price level, whereas its effect on compliance does not.

3.10. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 8) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize –31%, rice –27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of SALS over the rules does not grow with warming (Table A.16).

3.11. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop-site pairs, in at least one split), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr⁻¹

under the baseline climate and 16.0 PWh yr⁻¹ at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr⁻¹, and static arrays 6.4 and 6.9 PWh yr⁻¹. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply 16 PWh yr⁻¹ and the best 5% 40 PWh yr⁻¹ with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is probably too mild (Section 3.7), account for 53% of the electricity of the upscaling (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only wheat, rice and potato would give about 7 PWh yr⁻¹ at the same coverage, so the global figure should be read as an upper estimate for about half of its content. By crop, the LER of SALS under the baseline climate is 1.20 for wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

4. Discussion

4.1. What the results establish

Four findings stand on consistent evidence. First, sharing light with the crop is the main lever on the electricity that tracker-based agrivoltaic arrays can produce under a yield floor. With static trackers the floor limits arrays to about 6% ground coverage and an LER of 1.07. Any light-sharing control allows about three times the density, and the LER rises to 1.18–1.24 for handwritten rules. Second, the best simple rule follows crop phenology. It shares light across the middle of the season, around flowering and grain filling, and produces electricity at the beginning and the end. It needs only the crop’s thermal time, which is observable on the farm, and it is insensitive to forecast errors, whereas the rule that reacts to heat and drought stress is worse in

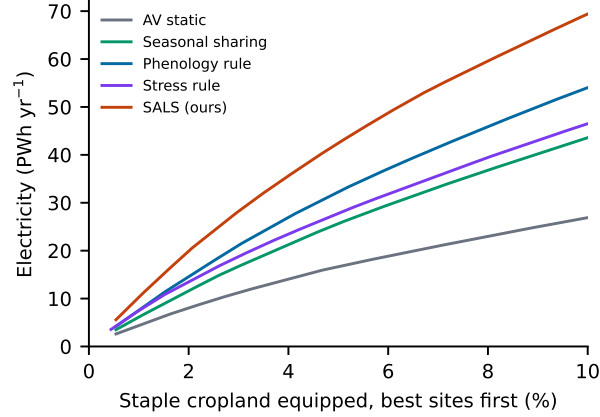


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

every climate and degrades with forecast error. Third, the qualitative result does not depend on the weakest part of the model. The simulated shade response is conservative against the field trials we could use (Section 3.6); calibrating it raises the allowed density, and the phenology rule stays the best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative, calibrated and field-fitted responses. However, the response learned from the meta-analysis shows steeper losses for maize and grain legumes than the simulator reproduces (Section 3.7), so the absolute gains are conservative for wheat, rice and potato and probably overestimates for maize and soybean (53% of the upscaled electricity). Fourth, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.04 in LER against the written rule (0.018 against a rule whose window was tuned on the training seasons, 5.7% more electricity at equal retention), reproducibly over training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four climates, and at 84–86% of the held-out sites. This gain is small compared with the gap between static and dynamic control.

4.2. What role does artificial intelligence play?

We tested three ways in which learning could matter, and the answers differ. The first is learning the crop’s response to shade from data. The learned

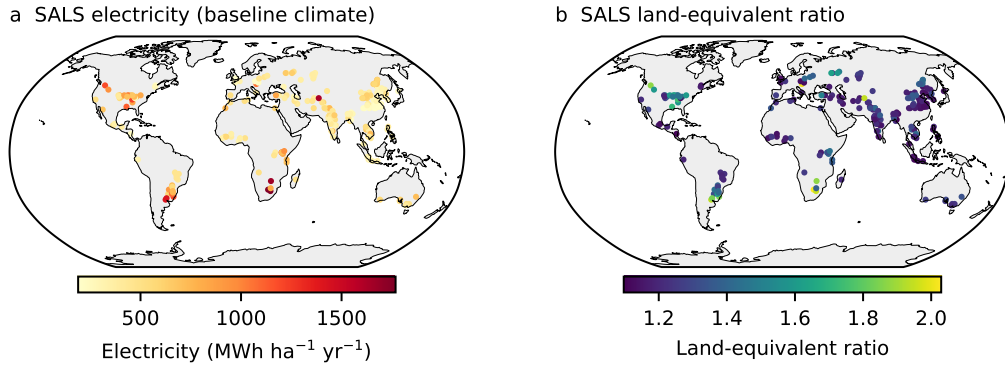


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

842 model is useful mainly as a measurement of how little can be learned: a hier-
843 archical model fitted to 428 observations from 58 studies predicts a held-out
844 study of the four staple crop types with an error of 0.22 in relative yield, no
845 better than proportional loss, and the gain over proportional loss (0.43 to
846 0.31 over all types) comes from crop types that the simulator does not repre-
847 sent. Crop type is the information that transfers, whereas irradiation, climate
848 zone and shade type add nothing measurable and a non-parametric model
849 (gradient-boosted trees) is worse with this number of studies (Section 3.7).
850 Two things follow. The shade response of the staple crops is a distribution
851 with a spread of about 20 percentage points between studies and not a curve,
852 and the process-based simulator does not reproduce the steep losses of maize
853 and legumes seen in the data. Learning the response was limited by the data
854 and not by the model: two studies support the tuber curve, no rice study
855 exists, and 46 panel observations are all leafy vegetables.

856 The second is calibration through the differentiable simulator. Gradient
857 descent through whole seasons fitted the response of each crop to the learned
858 curves and to the field trials, and it produced an ensemble of plausible re-
859 sponses that reconciles them (Section 3.8). The ensemble informs the design
860 decision: a density chosen under the mean response leaves about 70% of
861 crop–site pairs at the floor if another plausible response holds, against 85–
862 87% for an oracle, and a density chosen to meet the floor in 75% of plausible
863 responses recovers about half of that gap at a cost of 0.04 LER. A simple mar-
864 gin, however, does better: raising the floor from 0.90 to 0.92 under the mean

865 response reaches 85–89% of the pairs and keeps 0.02–0.04 more LER than
 866 the ensemble rule at equal compliance. What the ensemble adds is the size
 867 of the margin, not a better rule. The quantile rule is a scenario-type design
 868 (Calafiore and Campi, 2006; Campi and Garatti, 2008); with five members it
 869 has none of the finite-sample guarantees of that theory, and the comparison
 870 with the margin shows that it is not needed in practice. The bottleneck of
 871 tracker-based agrivoltaics under a food-security floor is the uncertainty of
 872 the crop response that enters the design, and it is data on that response, not
 873 larger networks, that would improve the result; the learned curves show a
 874 spread of about 20 percentage points between studies. The guarantee holds
 875 only within the range of responses that the ensemble spans.

876 The third is control, and here learning contributes little. The learned
 877 controller adds 0.018 LER over a tuned phenology rule (and 0.044 over the
 878 written one), and 0.015–0.038 over the written rule under the four alternative
 879 responses on one split (21 pairs); a reader who needs a simple, auditable
 880 controller can use that rule and obtain most of the electricity gain over static
 881 arrays. Neither of the two extensions that we expected to matter under
 882 uncertainty did: training on ensembles of responses (E1, E2 and E3) did not
 883 help and, for the wider ensembles, did not retain the margin over the written
 884 rule within our training budget (one seed, 21 pairs), and neither did giving
 885 the controller a farm-measurable signal, the biomass of the crop relative to
 886 a reference plot, nor a feedback rule built on that signal. The learned policy
 887 also has to be initialised from the phenology rule (Section 4.5). We therefore
 888 do not claim that learned control is what makes agrivoltaics work: in this
 889 model, the schedule that a learned policy converges to is a smooth function
 890 of crop phenology, which a rule can already express.

891 *4.3. What the results do not establish*

892 The early hypothesis that stress-aware control, that is, shading the crop
 893 on heat and drought days, would be a main source of benefit under warming
 894 was not supported. The stress rule performs worse than the phenology rule
 895 in every climate, and a learned controller without stress inputs performs as
 896 well as one with them. A plausible reason is that in our crop model the yield
 897 gain from shading on a hot or dry day is smaller than the loss of light on
 898 that day, so stress information does not help. Whether this holds for crop
 899 models with more detailed canopy-temperature and heat-stress responses is
 900 an open question. Agrivoltaics does not protect staple production against
 901 warming in our simulations: light-sharing systems produce about 90% of the

open-field yield in every climate and follow the same decline in maize and rice (Fig. 8).

4.4. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.11) shows the trade-off in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the trials (rice, wheat, potato, maize) but too mild for maize and soybean relative to the meta-analysis, where the 40%-shading relative yields are 0.42 and 0.40 against 0.75 and 0.48 in the simulator. The harvest-index sensitivities h are assumptions, and the sensitivity analysis uses rule controllers on a subsample of 22 sites.
- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2°C, but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.
- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency. More detailed models with grain-number formation and canopy energy balance may favour different schedules.

- 937 • *Climate scenarios.* Warming was applied as uniform temperature shifts
938 with fixed precipitation, radiation and relative humidity, and with un-
939 changed cultivars and sowing dates. It is not a bias-adjusted climate
940 projection.
- 941 • *Initialisation and state information.* The network is initialised from the
942 phenology rule, because training from random weights collapsed to full
943 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
944 and we cannot say whether a better optimiser or a richer policy class
945 would find larger gains. The controller uses the simulated state of the
946 crop and soil (thermal time, canopy interception, soil water) without
947 measurement error, and the forecast errors we tested apply to weather
948 only.
- 949 • *Economics and deployment.* The economic analysis (Section 3.9) uses
950 assumed prices and costs, no grid, curtailment, storage or subsidy
951 model, and does not value food security; its break-even prices move
952 with the capex and the discount rate (Fig. 7). The value of midday
953 electricity, curtailment, storage, structural cost and farmer acceptance
954 (Trommsdorff et al., 2021) will determine the realisable share. The av-
955 erage row pitch at the chosen densities (about 13 m for 2.2-m modules)
956 leaves room for machinery, but the densest designs (GCR up to 0.5)
957 have a pitch of only 4.4 m.

958 5. Conclusions

959 Light sharing, the daily choice between maximum electricity and letting
960 light through to the crop, is the main lever of tracker-based agrivoltaics un-
961 der a food-security floor. In a differentiable simulator of trackers, PV output,
962 crop light, microclimate and crop growth, evaluated fairly on 400 crop-site
963 pairs with the array density of each site chosen from its own history, light-
964 sharing rules allowed about three times the module density of static arrays
965 at a mean yield retention of 90%. They raised the land-equivalent ratio
966 from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no
967 weather forecast was the best of them. The simulated shade response is con-
968 servative against the field trials we could use for rice, wheat and potato, and
969 calibrating it raised the gains without changing the ranking of methods; for
970 maize and soybean, which supply 53% of the upscaled electricity, the meta-
971 analysis indicates larger losses than the simulator reproduces whereas one

maize trial indicates smaller ones, so the gains for these crops are uncertain and possibly overestimated. The crop's response to shade, the least certain input, was learned from 428 field observations: a hierarchical model predicted a held-out study of the staple crop types no better than proportional loss (error 0.22), which shows the size of the between-study spread, and gradient descent through the simulator turned it into an ensemble of plausible responses. A learned controller exceeded the written phenology rule by 0.044 LER and a rule with a window tuned on the training seasons by 0.018 (95% interval 0.012–0.024), by following a smooth phenology-dependent schedule that a rule can express; training it on ensembles of shade responses or giving it a farm-measurable biomass signal did not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (one seed, 21 pairs). The decision that mattered was the design. A density chosen under the mean response left about 70% of crop–site pairs at the floor when another plausible response held (oracle 85–87%); choosing the density so that the floor holds in 75% of the plausible responses recovered about half of the gap at a cost of 0.04 LER, but a floor raised by 0.02 under the mean response did better at equal compliance. The ensemble is therefore useful to size a safety margin, not to replace it. Under illustrative costs, light sharing breaks even with the open field at 73 (learned controller) to 92 USD MWh^{−1}; a conventional plant replacing the crop stays more profitable at every price, so agrivoltaics depends on a premium, land-use rules or a value for food production that markets do not give. The yield floor is met on average but not in every season or region, and the electricity potential of 5% of staple cropland (about 14 PWhyr^{−1}) requires about 10 TWp of modules. Future work should validate the crop response to shade under trackers with more field data, notably for maize and legumes where the meta-analysis and the trials disagree, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers that guarantee the food-security floor per site and per season.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

1006 **Declaration of competing interest**

1007 The author has no known competing financial interests or personal re-
1008 lationships that could have appeared to influence the work reported in this
1009 paper.

1010 **Data and code availability**

1011 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
1012 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
1013 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
1014 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
1015 ments, together with the site list, is available from the author’s public reposi-
1016 tory (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

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1020 **Appendix A. Statistical comparison**

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Rules and the learned controller pooled over the three spatial splits (76 distinct crop-site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05). The tuned phenology rule shares light while thermal time lies between 0.15 and 1.0 of the season requirement (window chosen on training-block seasons; the original rule uses 0.05 to 1.0).

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule	0.154	0.292	0.916	85	91	1.207
Tuned phenology rule	0.161	0.317	0.916	84	89	1.233
SALS	0.161	0.335	0.916	83	87	1.251

Table 5: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 6: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 7: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 8: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 9: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 10: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 11: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). Robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional (conservative) and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
SALS (trained on proportional response)	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 12: Array design under shade-response uncertainty, phenology rule, pooled over three spatial splits (76 distinct crop–site pairs; every third held-out pair). Truth: each of five members of the ensemble in turn; designs of the first group use the other four members (leave-one-out). Ensemble E2 is calibrated to the learned curves and field trials, E3 additionally carries the between-study spread (target errors 0.2). Pairs ≥ 0.9 : share whose mean retention reaches the floor. The last three rows keep the density rule of the main analysis but raise the floor used in the design.

Design	E2		E3	
	Pairs ≥ 0.9 (%)	LER	Pairs ≥ 0.9 (%)	LER
oracle (true response)	87	1.486	85	1.515
mean response	69	1.500	70	1.523
floor met in 50% of responses	65	1.510	67	1.541
floor met in 75% of responses	78	1.462	79	1.489
floor met in 90% of responses	89	1.412	88	1.409
floor 0.92 under mean response	89	1.434	85	1.468
floor 0.94 under mean response	98	1.351	93	1.407
floor 0.96 under mean response	99	1.258	98	1.313

Table 13: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 14: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 13). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity	Crop loss	Net at 60	Net at 90	Net at 120
AV static	200	11.5	0.27	0.3	6.3	12.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5	9.2
Stress rule	336	23.2	0.30	-3.3	6.8	16.9
Phenology rule	382	30.1	0.30	-7.5	4.0	15.4
Tuned phenology rule	415	31.6	0.31	-7.0	5.5	17.9
SALS	438	31.6	0.31	-5.6	7.6	20.7
Conventional PV plant	1306	58.7	3.60	16.0	55.2	94.4

Table 15: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. The capex of the conventional plant is 0.75/0.70 of that of the agrivoltaic plant in the capex rows.

Parameter	Value	Static arrays	Tuned rule	SALS	Conventional plant
Base		6.3	5.5	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	14.9	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-10.3	-8.2	25.8
Discount rate	4%	8.0	10.3	12.4	64.1
Discount rate	8%	4.4	0.2	2.4	45.4
Crop prices	0.5×	6.4	5.6	7.7	57.0
Crop prices	2.0×	6.0	5.2	7.3	51.6
Electricity price	60	0.3	-7.0	-5.6	16.0
Electricity price	120	12.3	17.9	20.7	94.4

Table A.16: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92